

The Complete Model Matrix (Plain English)

Parkinson's Disease Progression & Multimodal Deep Learning Capstone

Primary Target: MDS-UPDRS Part III (NP3TOT)
Prediction Horizons: +12 & +24 Months

STEP 1: THE YARDSTICKS
Baselines (M1 – M3)
Establish the foundation and prove that visit history and brain scans carry meaningful predictive signal.

STEP 2: THE SPECIALISTS
Deep Encoders (M4 & M5)
Model 4 specializes in symptoms over irregular hospital visits; Model 5 specializes in 3D brain measurements.

STEP 3: THE FUSION CONTEST
Multimodal Fusion (M6 – M8)
Compete 3 strategies (Early, Late, Dynamic Attention) to discover which produces the most stable 2-year predictions.

MODEL & CATEGORY	WHAT DATA GOES IN? (INPUT)	HOW IT WORKS (BRAIN OF THE MODEL)	WHAT COMES OUT?	WHY WE ARE BUILDING IT (PURPOSE)
Model 1: Snapshot Baseline BASELINE	Today's visit only Current symptoms, age, medication state (ON/OFF), Levodopa dose	Decision Trees (XGBoost) Splits data with simple if/then rules like a clinical flowchart.	Score at +12m & +24m	The baseline test: Proves what can be predicted from just a single doctor visit.
Model 2: History Baseline BASELINE	Full visit history All past clinical visits in chronological sequence	Standard Memory Network (LSTM) Remembers past visits, assuming visits happen at equal time intervals.	Score at +12m & +24m	Answers RQ1: Proves whether having patient visit history beats a single visit alone.
Model 3: Imaging Baseline BASELINE	Brain scan numbers only DaTSCAN dopamine binding (SBR) + FreeSurfer MRI brain volumes	Simple Neural Network Directly maps brain measurements to motor score severity.	Score at +12m & +24m	Modality test: Benchmarks what brain scans can predict alone before any fusion.
Model 4: Smart Time Model DEEP ENCODER	Visit history + exact days between visits Handles irregular intervals & missed check-ups	Decaying Memory Network (GRU-D) Fades older memories and handles gaps based on real elapsed days.	Patient Health Vector	Fixes real hospital flaws: Real patients don't visit on fixed intervals; GRU-D handles real-world gaps.
Model 5: Deep Brain Encoder DEEP ENCODER	All brain scan measurements DaTSCAN SBR + MRI subcortical brain volumes + cortical thickness	Deep Residual Neural Network Learns non-linear interactions across multiple brain structures.	Brain Health Vector	Feature compression: Compresses dozens of brain metrics into a clean summary for fusion.
Model 6: Early Fusion MULTIMODAL FUSION	Health Summary + Brain Summary glued side-by-side	Joint Neural Network Processes both summaries mixed together all at once from the beginning.	Score at +12m & +24m	Answers RQ2 (Option A): Tests whether simple feature concatenation is sufficient.

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Model 7: Late Fusion MULTIMODAL FUSION	Clinical prediction + Brain prediction made separately first	Smart Blending Judge Lets each modality make an independent vote, then learns optimal trust weights.	Score at +12m & +24m	Answers RQ2 (Option B): Tests whether independent voting prevents noise leakage between modalities.
Model 8: Attention Fusion ★ FLAGSHIP WINNER	Both summaries fed into an interactive Attention Gate	Dynamic Attention Network Acts like an expert doctor: dynamically decides whether to trust symptoms or brain scans per patient.	Score at +12m & +24m	Answers RQ2 (Flagship): Most advanced model; provides explainable attention weights for each patient.
Model 9: 3D Brain Scanner OPTIONAL STRETCH	Actual 3D MRI brain scans Raw voxel volume data	3D Computer Vision (3D DenseNet) Scans 3D brain images slice-by-slice.	3D Brain Features	Phase 4 Stretch: Tests if looking at raw brain images beats pre-calculated volume numbers.
Model 10: Smartphone Model OPTIONAL STRETCH	mPower phone sensor data Voice recordings, finger tapping, walking steps	Audio & Motion Network Removes microphone noise and extracts digital tremor signals.	Phone Health Vector	RQ3 Stretch: Tests if noisy crowdsourced phone data brings signal beyond in-hospital tests.